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SSIG: A Visually-Guided Graph Edit Distance for Floor Plan Similarity

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Abstract

*We propose a simple yet effective metric that measures structural similarity between visual instances of architectural floor plans, without the need for learning. Qualitatively, our experiments show that the retrieval results are similar to deeply learned methods. Effectively comparing instances of floor plan data is paramount to the success of machine understanding of floor plan data, including the assessment of floor plan generative models and floor plan recommendation systems. Comparing visual floor plan images goes beyond a sole pixel-wise visual examination and is crucially about similarities and differences in the shapes and relations between subdivisions that compose the layout. Currently, deep metric learning approaches are used to learn a pair-wise vector representation space that closely mimics the structural similarity, in which the models are trained on similarity labels that are obtained by Intersection-over-Union (IoU). To compensate for the lack of structural awareness in IoU, graph-based approaches such as Graph Matching Networks (GMNs) are used, which require pairwise inference for comparing data instances, making GMNs less practical for retrieval applications. In this paper, an effective evaluation metric for judging the structural similarity of floor plans, coined **SSIG** (**S**tructural **S**imilarity by **I**oU and **GED**), is proposed based on both image and graph distances. In addition, an efficient algorithm is developed that uses SSIG to rank a large-scale floor plan database. Code will be openly available.*

1. Introduction

Floor plans, as one of the most celebrated media for communicating and thinking in architectural design, are horizontal orthographic projections of a particular building's floor. Floor plans are simple but powerful representations of space, conveying a richness of information about the compositional structure of buildings – the shapes, proportions, and relations between the building's individual subdivisions. The compositional structure of a building is a key factor in the architectural quality.

The multi-modal nature of floor plans as pictorial (as images), graphical (as a set of geometrical entities), and topological (as graphs) entities, leverage different data modeling and machine learning frameworks for digital representation and machine understanding in the domain of architectural design. In particular, floor plan synthesis, such as floor plan generation [9, 10, 18] or style transfer [1], has often been explored. Another major research line is that of architectural element recognition in which the goal is to automatically learn to extract geometrical features from floor plan images *e.g*. walls or furniture [7, 3, 14].

Despite the developments in the generative models for floor plans, there have been few valuable attempts at developing data-driven methods to *reason* about floor plans, such as floor plan retrieval by effectively comparing data instances *i.e*. measuring similarity. As found by [13], structural similarity judgment for floor plans is a challenging task – it is multi-faceted and is about the shapes and proportions, visual features, and relations between the floor plan's subdivisions. Generally speaking, comparing floor plans is about finding commonalities in the underlying structure, hence referred to as *structural similarity*.

The most effective methods for measuring structural similarity between floor plans rely on deeply learned vector representations with the goal to make the feature vector space reflect structural similarity *i.e*. the distance between feature vectors is small for structurally similar floor plans and vice versa. To train such models, [8] propose to use weakly supervised deep metric learning setup by obtaining similarity labels through Intersection-over-Union (IoU). To compensate for the lack of structural awareness in IoU, [13, 4] use Graph Matching Networks (GMNs) [6] that naturally embed topological information in the feature vectors. Besides the fact that GMNs hinder the practical usability [22], we believe that simpler frameworks could learn similarly effective vector representations when the objective is better aligned with structural similarity in the first place. In our work, we develop an improved objective and opt for simplicity. We start by investigating the merits and bottlenecks of well-known image- and graph-based similarity metrics.

The *Intersection-over-Union* (IoU) is often used as an evaluation metric to measure structural similarity *e.g*. to

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SSIG：一种用于平面布局相似性度量的视觉引导图编辑距离

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摘要

*We propose a simple yet effective metric that measures structural similarity between visual instances of architectural floor plans, without the need for learning. Qualitatively, our experiments show that the retrieval results are similar to deeply learned methods. Effectively comparing instances of floor plan data is paramount to the success of machine understanding of floor plan data, including the assessment of floor plan generative models and floor plan recommendation systems. Comparing visual floor plan images goes beyond a sole pixel-wise visual examination and is crucially about similarities and differences in the shapes and relations between subdivisions that compose the layout. Currently, deep metric learning approaches are used to learn a pair-wise vector representation space that closely mimics the structural similarity, in which the models are trained on similarity labels that are obtained by Intersection-over-Union (IoU). To compensate for the lack of structural awareness in IoU, graph-based approaches such as Graph Matching Networks (GMNs) are used, which require pairwise inference for comparing data instances, making GMNs less practical for retrieval applications. In this paper, an effective evaluation metric for judging the structural similarity of floor plans, coined **SSIG** (**S**tructural **S**imilarity by **I**oU and **GED**), is proposed based on both image and graph distances. In addition, an efficient algorithm is developed that uses SSIG to rank a large-scale floor plan database. Code will be openly available.*

1. 引言

平面图作为建筑设计中最为重要的表达与思维媒介之一，是特定建筑楼层的水平正投影。平面图是一种简洁而有力的空间表征方式，能够传达关于建筑构成结构的丰富信息——包括各功能分区的形态、比例及其相互关系。建筑的构成结构是决定其建筑品质的关键因素。

平面图的多模态特性表现为图像形式（作为…年龄）、图形（作为一组几何实体）以及拓扑（作为图结构）等实体，在建筑设计领域中分别利用不同的数据建模与机器学习框架实现数字化表征与机器理解。其中，平面布局的综合生成，例如平面布局的自动生成[9, 10, 18]或风格迁移[1]，一直是研究的重点。另一条重要的研究主线是建筑构件识别，其目标在于自动从平面布局图像中提取几何特征，如墙体或家具[7, 3, 14]。

尽管用于平面布局的生成模型取得了进展，但鲜有有价值的尝试致力于开发数据驱动的方法来对平面布局进行*reason*，例如通过有效比较数据实例*i.e*来实现平面布局检索。测量相似性方面亦是如此。正如[13]所指出的，平面布局的结构相似性判断是一项具有挑战性的任务——它具有多维度特征，涉及形状与比例、视觉特征以及平面布局各分区之间的关系。广义而言，平面布局的比较旨在识别其底层结构中的共性，因此被称为*structural similarity*。

用于度量楼层平面之间结构相似性的最有效方法，依赖于深度学习得到的向量表示，其目标是使特征向量空间能够反映结构相似性*i.e*。对于结构上相似的楼层平面，其特征向量之间的距离较小；反之亦然。为训练此类模型，文献[8]提出采用弱监督的深度度量学习框架，并通过交并比（IoU）获取相似性标签。为弥补IoU在结构感知方面的不足，文献[13, 4]则采用图匹配网络（GMN）[6]，该方法能够自然地将拓扑信息嵌入到特征向量中。然而，除GMN在实际应用中存在一定的局限性[22]之外，我们认为，若能首先使优化目标与结构相似性更好地对齐，更为简单的框架同样可以学习到同样有效的向量表示。在本研究中，我们提出了一个改进的优化目标，并坚持简洁的设计原则。我们首先系统地考察了现有基于图像和基于图的相似性度量方法的优点与瓶颈。

Intersection-over-Union (IoU)通常被用作一种评估指标，用于衡量结构相似性*e.g*。

find similar or matching pairs for deep metric learning models [13, 4], to evaluate the accuracy of structural reasoning algorithms [7], or to evaluate the output diversity of generative models [9]. In this paper, we show how IoU fails to robustly measure structural similarity because 1) IoU is overly sensitive to geometric perturbations in the image representation of a floorplan and 2) IoU is often unable to identify important differences in the connectivity between space subdivisions such as doors, walls, openings.

Floor plans have been successfully modelled as graphs [9, 13], which allows for graph matching algorithms to compute a distance (similarity) between a pair of graphs. For example, the *Graph Edit Distance* (GED) computes the minimum cost of converting a source graph into another (isomorphic to it) target graph [17]. For example, [9] uses GED to check the compatibility between generated layouts. Although GED effectively captures topological similarity and with that addresses some of IoU's limitations, we show that GED has inherent limitations when used for measuring structural similarity.

Independently, image- and graph-based similarity metrics lack to holistically capture the structural similarity between floor plans. Therefore, instead of treating images and graphs in isolation, we propose an evaluation metric that ascertains closeness based on the floor plans' corresponding image *and* graph, called **SSIG**. In addition, an efficient algorithm based on **SSIG** is used to rank RPLAN [26], a large-scale floor plan database, on structural similarity. Our contributions include:

- A study on the distributions and correlations between IoU and GED on a large-scale floor plan database, with one main finding: IoU and GED independently fail to robustly capture structural similarity.
- A proposed simple yet powerful measure for the structural similarity of floor plans.
- An effective ranking technique to sort a large floor plan database, hence developing a starting point for proper evaluation and training of floor plan retrieval models.

2. Related works

Floor plan representations. Floor plans are digitally represented in various ways that each emphasize different components of interest *e.g.* as images allowing to model fine-grained details such as materialization and furniture [21, 5], geometries that explicitly model the shapes, proportions and locations of the shapes of the subdivisions [13], and graphs that model the relations between the subdivisions [9, 13, 18, 12].

Floor plan analysis. Traditional approaches on floor plan analysis – or similar data such as documents [28] or UIs [2] – involve primitive heuristics to approximate a floor plan distance and are, hence, instance-specific [11]. More recently, two-step approaches became the standard to approximate similarity of floor plans [27, 16, 20, 23, 21, 19, 25, 12], in which the first step involves extracting relevant features from an image *e.g.* a graph representation of the room connectivity [12] and the second step computes a distance based on the extracted features *e.g.* a subgraph matching algorithm for graph similarity [16]. Besides the fact that the two-step approaches are prone to error propagation, the two-step approaches assume that the extracted features, such as the access graph, completely and accurately capture all aspects of floor plans. In our work, we challenge this assumption, particularly concerning semantic image and access graph representations of floor plans.

Structural similarity by Graph Neural Networks. Recently, Graph Neural Networks (GNN) have been proposed to *learn* floor plan similarity [13, 8, 4], learning floor plan vector representations in an end-to-end fashion. Most successfully, [4, 13] leverage Graph Matching Networks (GMN) [6] in combination with weakly-supervised labels based on the IoU. Due to the cross-graph information sharing in GMNs, [13] found that GMNs compensate for the lack of structural awareness in IoU and naturally learn to embed the structural commonalities in the vector representations. A critical shortcoming of GMNs is that vector representations cannot be computed in isolation, therefore limiting the practical usability. We believe that the main reason that cross-graph information sharing is needed is that essentially the wrong objective is injected – that is to mimic a distance metric inspired by IoU. We show that our metric, which is a combination between visual- and a graph-inspired similarity scores, can be leveraged to rank a floor plan database for which the retrievals share similar characteristics as to the works in [13].

3. Method

In this section, we develop a simple yet effective strategy to numerically approximate structural similarity between floor plans. Formally, we seek a *similarity function* $s(\cdot,\cdot)$ between two floor plans p_1 and p_2:

$$s(p_1, p_2) : \mathcal{P} \times \mathcal{P} \rightarrow \mathbb{R}^+, \tag{1}$$

in which $\mathcal{P}$ is the (mathematical) space of a floor plan. $\mathcal{P}$ can be multi-modal *e.g.* a joint space of images and graphs: $\mathcal{P} = \mathbb{R}^{H \times W \times 3} \times \mathcal{G}$. The goal is to make s closely mimic the structural similarity between floor plans: the larger the structural similarity the larger s and vice versa.

The section is split into three main parts. The first part explores the modeling and attribution of floor plans, (Subsec. 3.1). The second part investigates known and

用于为深度度量学习模型寻找相似或匹配的样本对[13, 4]，评估结构推理算法的准确性[7]，或评价生成模型的输出多样性[9]。在本文中，我们指出交并比（IoU）无法稳健地衡量结构相似性，原因在于：1）IoU 对平面布局图像表示中的几何扰动过于敏感；2）IoU 往往难以识别空间细分之间（如门、墙、洞口等）的重要连通性差异。

平面布局已成功被建模为图结构[9, 13]，从而使得图匹配算法能够计算一对图之间的距离（相似度）。例如，*Graph Edit Distance* (GED)用于计算将源图转换为另一张与其同构的目标图所需的最小代价[17]。以文献[9]为例，该研究利用GED来检验生成布局的兼容性。尽管GED能够有效刻画拓扑相似性，并在一定程度上克服了IoU的一些局限性，但我们表明，在用于度量结构相似性时，GED本身也存在固有的局限性。

独立来看，基于图像和图的相似性度量均难以全面捕捉平面布局之间的结构相似性。因此，我们并未将图像与图分别处理，而是提出了一种评估指标——SSIG，该指标基于平面布局对应的图像-图对来判定其相似程度。此外，我们还基于SSIG设计了一种高效算法，用于对RPLAN（一个大规模平面布局数据库）按照结构相似性进行排序。我们的主要贡献包括：

- 一项针对大规模平面布局数据库中IoU与GED分布及其相关性的研究，其主要发现为：IoU与GED均无法独立且稳健地表征结构相似性。
- 一种用于衡量楼层平面结构相似性的简单而有效的指标。
- 一种有效的排序技术，用于对大型平面布局数据库进行排序，从而为平面布局检索模型的合理评估与训练提供初始依据。

2. 相关工作

平面图的表示方法。平面图在数字化表示中存在多种形式，每种形式均侧重于不同的关注要素*e.g*：以图像形式呈现，能够刻画诸如材质和家具等细粒度细节[21, 5]；以几何体形式呈现，可显式建模各功能分区的形状、比例及其空间位置[13]；以及以图结构形式呈现，用于建模各功能分区之间的关联关系[9, 13, 18, 12]。

楼层平面分析。针对楼层平面分析的传统方法——或类似的数据，如文档[28]或用户界面——[2]——采用原始启发式方法来近似计算平面布局之间的距离，因此具有实例特定性[11]。近年来，两步法已成为近似计算平面布局相似性的标准方法[27, 16, 20, 23, 21, 19, 25, 12]，其中第一步从图像中提取相关特征（如v7），构建房间连通性的图表示[12]；第二步则基于所提取的特征（如v8）计算距离，并采用子图匹配算法来度量图的相似性[16]。然而，除了两步法容易产生误差传播之外，这类方法还隐含了一个假设，即所提取的特征（如访问图）能够完整且准确地刻画平面布局的所有方面。在本研究中，我们对这一假设提出了质疑，尤其是在涉及平面布局的语义图像表示和访问图表示时。

基于图神经网络的结构相似性。近期，图神经网络（GNN）已被用于表征楼层平面的相似性[13, 8, 4]，并以端到端的方式学习楼层平面的向量表示。其中，文献[4, 13]成功地将图匹配网络（GMN）[6]与基于交并比（IoU）的弱监督标签相结合。由于GMN中存在跨图的信息共享机制，文献[13]发现，GMN能够弥补IoU在结构感知方面的不足，并自然地学习将结构共性嵌入到向量表示中。然而，GMN的一个关键缺陷在于其向量表示无法独立计算，从而限制了实际应用的可行性。我们认为，之所以需要跨图信息共享，根本原因在于所采用的目标函数本质上存在偏差——即试图模仿一种受IoU启发的距离度量。我们证明，我们的度量方法通过融合视觉特征与图结构特征的相似性得分，可用于对楼层平面数据库进行排序，且检索结果与文献[13]中的工作具有相似的特性。

3. 方法

在本节中，我们提出了一种简单而有效的策略，用于数值近似楼层平面之间的结构相似性。形式上，我们寻求两个楼层平面p_1和p_2之间的映射关系 *similarity function* $s\left(\cdot,\cdot\right)$：

$$s\left(p_1, p_2\right) : \mathcal{P} \times \mathcal{P} \rightarrow \mathbb{R}^{+}, \tag{1}$$

其中，$\mathcal{P}$ 为平面布局的（数学）空间；$\mathcal{P}$ 可以是多模态的 *e.g*，即图像与图结构的联合空间：$\mathcal{P} = \mathbb{R}^{H\times W\times 3} \times \mathcal{G}$。目标是使 s 尽可能地逼近平面布局之间的结构相似性：结构相似性越大，s 越大，反之亦然。

该部分分为三个主要部分。第一部分探讨了平面图的建模与归因问题（第3.1节）。第二部分则研究了已知的以及

frequently used image- and graph-based similarity metrics (Subsec. 3.2, 3.3). The third part describes the new similarity metric (Subsec. 3.4), and provides an algorithm that can efficiently rank a floor plan dataset (Subsec. 3.5).

3.1. Floor plan representations

In this work, we consider three well-known floor plan representations: 1) the pictorial image, 2) the semantic image, and 3) the access graph.



Figure 1. **Floor plan representations**. Left: Pictorial image. Center: Semantic image. Right: Access graph.

Pictorial image Conventionally, floor plans are modeled as images. All floor plan images that are either grayscale or RGB and do not contain any categorial information at pixel level about the semantics of the subdivisions are referred to as *pictorial images*, $I \in \mathcal{I} = \mathbb{R}^{H \times W (\times 3)}$ (Fig. 1, Left).

Semantic image Floor plans are *structured arrangements* consisting of subdivisions – areas such as a living room or bathroom, openings such as doors or windows, and separators such as walls or railings. All floor plans that have categorical labels for subdivision areas and/or semantic information encoded at pixel level are referred to as *semantic images* in our definition, $X \in \mathcal{X} = \{0, 1, \ldots, N_c\}^{H \times W}$, in which N_c is the number of subdivision categories (Fig. 1, Center). In the semantic image representation, pixel values correspond to subdivision categories *e.g.* $1 \leftarrow$ "living room", $2 \leftarrow$ "bathroom", $16 \leftarrow$ "internal wall", etc.

Access graph Floor plans often have clearly identifiable *relations* between subdivisions *e.g.* access information between two rooms by means of a door. Even though semantic images efficiently capture the shapes and categories of the subdivisions, semantic image maps do not explicitly define the relations between the subdivisions.

Instead, floor plans can be attributed an *access graph*, directly laying bare the topological structure of the subdivisions (Fig. 1, Right). Formally, a graph is defined as a 4-tuple $g = (N, E, \mu, \nu)$, in which N is a finite set of nodes, E is the set of edges, and μ and ν are the node and edge labelling functions respectively [15]. The nodes in an access graph correspond to the areas, and the edges to connections between the areas.

Depending on the use-case and feature accessibility, the nodes and edges can have (one or several) attributes. Usually, room-type information is encoded on the nodes, and edges are only present if a door is in between two rooms, hence edges do not have an attribute [9, 18, 24]. Even though we agree that access connectivity is the most important relation between rooms, an important relation is disregarded: room adjacency. Room adjacency is an influential factor *e.g.* for decisions around privacy, structural integrity, and function. Access graphs, in the remainder of this work, have therefor a connectivity-type attribute, which is either "door" or "adjacent".

Proposed by [8], it is noteworthy to mention that several approaches *compute* the edge features. Specifically, edge features between two nodes are computed by pairwise geometric features, such as the ratio between areas or the relative position.

3.2. Intersection-over-Union

The *Intersection-over-Union* (IoU) is not considered a proper distance metric, but it is a well-adopted evaluation metric for tasks where measuring the overlap between regions or sets is important, such as in object detection or segmentation tasks. In turn, IoU is often used as a measure for estimating structural similarity in various applications related to floor plans *e.g.* for finding similar or matching floor plan pairs for similarity learning [13, 4], evaluating the accuracy of segmentation algorithms [7], or to check the diversity of outputs in floor plan generation [9]. The IoU is commonly defined as the fraction between the amount of overlap and the union of two binary images X_1 and $X_2 \in \{0, 1\}^{H \times W}$:

$$\text{IoU}\left(X_1, X_2\right) = \frac{X_1 \cap X_2}{X_1 \cup X_2}. \tag{2}$$

When more classes are present in an image *e.g.* in the case of semantic images ($X_1, X_2 \in \mathcal{X}$), IoU is computed per class and the results are aggregated to find the final score. For example, the mean IoU (mIoU) computes the average IoU, with respect to the ground truth, across the different classes, therefore treating every class similarly regardless of its proportion to the image size.

$$\text{mIOU}\left(X_1, X_2\right) = \frac{1}{N_c} \sum_{c=1}^{N_c} \text{IoU}\left(X_1 == c, X_2 == c\right), \tag{3}$$

where N_c is the number of classes.

Even though IoU is a powerful metric that in part is suitable for approximating structural similarity between floor plans, IoU frequently fails on both extremes: 1) false-negative: low IoU - high structural similarity, and 2) false-positive: high IoU - low structural similarity. 1) IoU is oversensitive to irrelevant geometric perturbations in the image

常用的基于图像和图的相似性度量指标（第3.2节、第3.3节）。第三部分介绍了新的相似性度量指标（第3.4节），并给出了一种能够高效对平面布局数据集进行排序的算法（第3.5节）。

3.1 地面平面图表示

在本研究中，我们考虑了三种广为人知的平面布局表示方法：1）图像表示，2）语义图像表示，以及 3）访问图表示。



图1. 平面布局表示。左：图像视图。中：语义图像。右：访问图。

图像表示 传统上，楼层平面图被建模为图像。所有既为灰度图像或RGB图像，且在像素级别上不包含关于各区域语义的类别信息的楼层平面图，均被称为*pictorial images*、$I \in \mathcal{I} = \mathbb{R}^{H \times W (\times 3)}$（（见图1，左侧）和)。

语义图像中的平面图是由若干子区域（如客厅、浴室等）、开口（如门、窗等）以及分隔构件（如墙体、栏杆等）组成的*structured arrangements*。在我们的定义中，所有为子区域赋予类别标签并以像素级编码语义信息的平面图均被称为*semantic images*，记作 $X \in \mathcal{X} = \{0, 1, \ldots, N_c\}^{H \times W}$，其中$N_c$表示子区域类别的数量（见图1，中心部分）。在语义图像表示中，像素值对应于各子区域类别，例如1代表“客厅”，2代表“浴室”，16代表“内墙”等。

访问图：楼层平面图通常在各功能分区之间存在明确可识别的*relations*，即通过门实现的两个房间之间的访问关系。尽管语义图像能够有效地表征各功能分区的形状与类别，但语义图像地图并未显式地定义各功能分区之间的关系。

相反，楼层平面图可以被赋予一个*access graph*，从而直接揭示各分区的拓扑结构（图1，右）。形式上，图被定义为一个四元组$g = (N, E, \mu, \nu)$，其中N为有限的节点集，E为边集，而μ和ν分别为节点与边的标注函数[15]。在接入图中，节点对应于各个区域，边则表示区域之间的连接。

根据用例和功能的可访问性，节点和边可以具有（一个或多个）属性。通常，房间类型信息编码于节点上，而只有当两间房间之间存在门时才会出现边，因此边本身不携带属性[9, 18, 24]。尽管我们认同通行连通性是房间之间最重要的关系，但仍有另一项重要关系被忽视：房间的邻接关系。房间的邻接关系是影响隐私、结构完整性及功能布局等决策的重要因素*e.g*。因此，在本文后续讨论中，通行图引入了一种连通性类型的属性，其取值为“门”或“相邻”。

由[8]提出，值得注意的是，现有若干方法会利用边特征。具体而言，两个节点之间的边特征是通过成对的几何特征来计算的，例如面积之比或相对位置等。

3.2. 交并比

Intersection-over-Union (IoU) 并不被视为一种严格意义上的距离度量，但在需要衡量区域或集合之间重叠程度的任务中，它是一种被广泛采用的评估指标，例如目标检测或分割任务。相应地，IoU 也常被用作一种度量方法，用于在与平面布局相关的各类应用中评估结构相似性，如用于相似性学习中的相似或匹配平面布局对的查找[13, 4]、分割算法精度的评价[7]，或用于检验平面布局生成结果的多样性[9]。IoU 通常被定义为两个二值图像 X_1 和 X_2 的交集面积与并集面积之比 $\in \{0, 1\}^{H \times W}$:

$$\mathrm{IoU}\left(X_1, X_2\right) = \frac{X_1 \cap X_2}{X_1 \cup X_2}. \tag{2}$$

当图像中存在更多类别时*e.g*，对于语义分割图像（$X_1, X_2 \in \mathcal{X}$），逐类计算交并比（IoU），并将各分类的计算结果进行聚合以得到最终得分。例如，平均交并比（mIoU）即在不同类别上分别计算相对于真实标签的交并比，并取其平均值，从而对每个类别给予同等权重，而不考虑其在图像中所占的比例。

$$\mathrm{mIOU}\left(X_1, X_2\right) = \frac{1}{N_c} \sum_{c=1}^{N_c} \mathrm{IoU}\left(X_1 == c, X_2 == c\right), \tag{3}$$

其中，N_c 表示类别数。

尽管交并比（IoU）是一种强大的度量指标，能够在一定程度上用于近似楼层平面图之间的结构相似性，但其在两个极端情况下往往失效：1）假阴性：低IoU—高结构相似性；2）假阳性：高IoU—低结构相似性。1）IoU对图像中无关的几何扰动过于敏感。

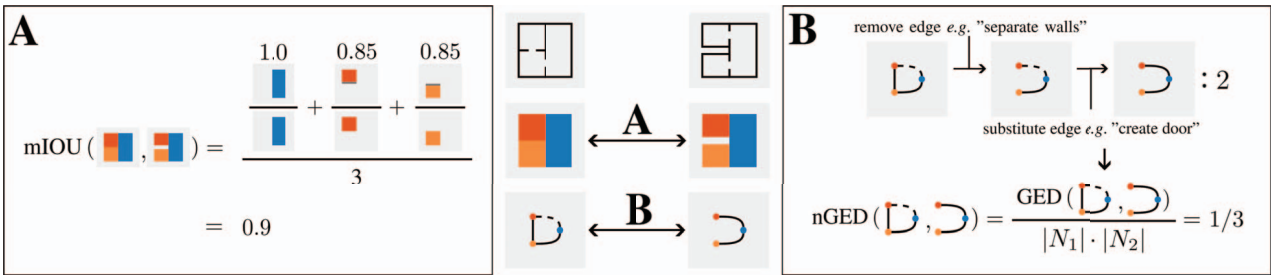


Figure 2. **Visual- and graph-based similarity metrics for floor plans**. A) IoU can be directly used as a measure for similarity between semantic image representations. mIoU between the semantic images is computed through Eq. 3 (floor plans have 3 semantic labels); in this case being 0.90 which is a relatively high number. B) nGED is computed through Eqs. 5 and 6: GED equals the shortest edit path which is 2; the orders of the graphs are both 3, hence nGED $= 2/9$. When $\gamma = 0.4$ (a typical value), $1 - \text{nGED}\,(g_1, g_2)^{\gamma} \approx 0.45$. The visual-based similarity (0.90) and graph-based (0.45) similarity scores are significantly different which indicates that in some cases IoU and GED negatively correlate in "seeing" similarity.

e.g. translation, rotations, and scale. For example, when identical floor plans are centered at different locations in an image, the IoU will be low; falsely indicating that the floor plans are *not* structurally similar. 2) IoU frequently fails to correctly identify the connectivity in floor plans *e.g*. small "air gaps" between rooms are hardly identified by IoU yet are indicative of structural characteristics (see example in Fig. 2).

3.3. Graph Edit Distance

The *Graph Edit Distance* (GED) [17], is another measure to judge similarity between floor plans, used for *e.g*. compatibility of generated layouts [10] or for floor plan retrieval [12]. GED is a metric that quantifies the similarity (dissimilarity) between two graphs by computing the minimum cost of transforming one graph g_1 into (a graph isomorphic to) g_2 in terms of a sequence of edit operations $(e_1, e_2, \ldots, e_j)$, referred to as the *edit path*. Edit operations typically include node and edge deletions, insertions, and substitutions, which can have different costs. Given a set of graph edit operations e_i and associated costs $c_i\,(e_i)$, the GED (g_1, g_2) is formally defined as,

$$\text{GED}\,(g_1, g_2) = \min_{(e_1, e_2, \ldots, e_j) \in \pi(g_1, g_2)} \sum_{j=1}^{k} c_j\,(e_j)\,, \tag{4}$$

in which $\pi\,(g_1, g_2)$ denotes the set of edit paths that transforms g_1 into (a graph isomorphic to) g_2. In our work, all edit operations have the same cost, hence Eq. 4 becomes,

$$\text{GED}\,(g_1, g_2) = \min_{k}\,(e_1, e_2, \ldots, e_k)\,, \tag{5}$$

such that $(e_1, e_2, \ldots, e_k) \in \pi\,(g_1, g_2)$. GED in this case is equivalent to the smallest number of edit operations. To normalize the GED between 0 and 1, we define the *Normalized Graph Edit Distance* (nGED) by dividing GED with the product of the orders[1] – the number of nodes – of the graphs ($|N_1|$ and $|N_2|$):

$$\text{nGED}\,(g_1, g_2) = \frac{\text{GED}\,(g_1, g_2)}{|N_1| \cdot |N_2|}. \tag{6}$$

nGED can be used as of measure of closeness between the topology of two floor plans, in which a small nGED suggests that two floor plans share topological characteristics, which could be, for example, indicative of commonalities in flow or function between two buildings.

nGED has several limitations. First of all, the amount of different floor plans possible for a particular access graph is huge, for which many corresponding floor plans pairs are not structurally similar (see Fig. 3). Second, it is unclear which edit costs align with structural similarity: Is the removal of a node or an edge more costly? Does the node label matter? etc. Third, nGED is compute-heavy which limits its use in real-time systems *e.g*. floor plan retrieval in search engines. We elaborate more on the shortcomings in Subsec. 4.2.

In Subsec. 4.3 we investigate the trends and correlations between nGED and mIoU on RPLAN [26] empirically.

3.4. A Visually-guided Graph Edit Distance

Independently, IoU and GED fail to holistically capture structural similarity between floor plans. Therefore, instead of treating images and graphs separately, we propose a metric that ascertains closeness based on the semantic image *and* access graph. We define the similarity between a pair of floor plans (p_1, p_2), $p_i = (X_i, g_i) \in \mathcal{X} \times \mathcal{G}$, as

[1] Note that Eq. 6 does not strictly normalizes GED. Even so, for floor plans *e.g*., in [26], nGED will always be below 1. Furthermore, the product of the orders, instead of for example the sum, nicely spreads the nGED distributions.

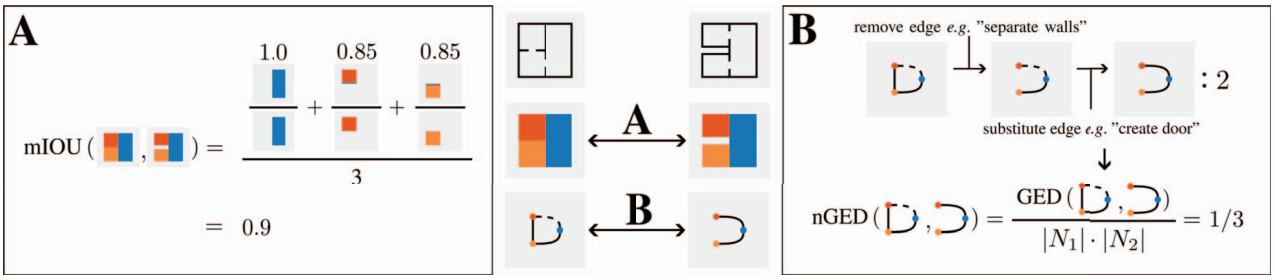


图2. 用于平面布局的视觉与图论相似性度量。A) IoU可直接用作语义图像表示之间相似性的度量指标。语义图像之间的mIoU通过式(3)计算得出（平面布局包含3个语义标签）；在本例中其值为0.90，属于较高水平。B) nGED通过式(5)和式(6)计算：GED等于最短编辑路径，其值为2；两图的阶数均为3，因此nGED为=2/9。当γ =0.4（典型取值）时，1−nGED$(g_1, g_2)^{\gamma}$ ≈0.45。基于视觉的相似性得分（0.90）与基于图论的相似性得分（0.45）存在显著差异，这表明在某些情况下，IoU与GED在“感知”相似性方面呈现负相关关系。

*e.g*平移、旋转和尺度变化。例如，当相同的平面图在图像中位于不同位置时，其交并比（IoU）会较低，从而错误地表明这些平面图在结构上是相似的。2）IoU经常无法正确识别平面图中的连通性。房间之间存在的微小“空隙”几乎无法被 IoU 识别，但这些空隙却能反映结构特征（参见图 2 中的示例）。

3.3 图编辑距离

Graph Edit Distance (GED) [17]是另一种用于度量楼层平面之间相似性的指标，常被应用于生成布局的*e.g*兼容性评估[10]或楼层平面检索[12]。GED是一种通过计算将一个图g_1转化为（与之同构的）图g_2所需的最小代价来量化两者之间相似性（或差异性）的度量，该代价以一系列编辑操作$e_1, e_2, \ldots, e_j$为单位，这些编辑操作被称为*edit path*。典型的编辑操作包括节点和边的删除、插入及替换，且每种操作可赋予不同的代价。给定一组图编辑操作e_i及其相应的代价$c_i(e_i)$，GED（g_1, g_2）的形式化定义为：

$$\mathrm{GED}(g_1, g_2) = \min_{(e_1, e_2, \ldots, e_j) \in \pi(g_1, g_2)} \sum_{j=1}^{k} c_j(e_j), \quad (4)$$

其中，$\pi(g_1, g_2)$ 表示将 g_1 转换为（与之图同构的）g_2 的所有编辑路径集合。在本研究中，所有编辑操作的代价均相同，因此式（4）可简化为：

$$\mathrm{GED}(g_1, g_2) = \min_{k} (e_1, e_2, \ldots, e_k), \quad (5)$$

使得 $(e_1, e_2, \ldots, e_k) \in \pi(g_1, g_2)$。在本例中，GED 等价于所需的最小编辑操作次数。为了将 GED 归一化至 0 到 1 之间，我们通过将 GED 除以……来定义 *Normal- ized Graph Edit Distance* (nGED)。

图$|N_1|$和图$|N_2|$的阶（即节点数）之积：

$$\mathrm{nGED}(g_1, g_2) = \frac{\mathrm{GED}(g_1, g_2)}{|N_1| \cdot |N_2|}. \quad (6)$$

nGED可用作度量两个楼层平面拓扑结构之间相似程度的指标；其中，较小的nGED值表明这两个楼层平面具有相似的拓扑特征，这可能反映出两栋建筑在人流或功能布局方面存在共性。

nGED 存在若干局限性。首先，对于给定的接入图而言，可能存在的不同平面布局数量极其庞大，其中许多对应的平面布局对在结构上并不相似（见图 3）。其次，尚不清楚哪些编辑代价能够准确反映结构相似性：移除一个节点或一条边的代价更高？节点标签是否具有重要性？等等。第三，nGED 的计算开销极大，这限制了其在实时系统中的应用*e.g*以及在搜索引擎中的平面布局检索。我们在第 4.2 节中将进一步详述这些不足之处。

在第4.3节中，我们基于RPLAN [26]数据集，对nGED与mIoU之间的趋势及相关性进行了实证研究。

3.4. 一种基于视觉引导的图编辑距离

独立来看，IoU 和 GED 均无法全面地刻画平面布局之间的结构相似性。因此，我们不将图像与图分别处理，而是提出一种基于语义图像访问图来度量其接近程度的指标。我们将一对平面布局（p_1, p_2）之间的相似性定义为：

[1]Note that Eq. 6 does not strictly normalizes GED. Even so, for floor plans *e.g*., in [26], nGED will always be below 1. Furthermore, the product of the orders, instead of for example the sum, nicely spreads the nGED distributions.

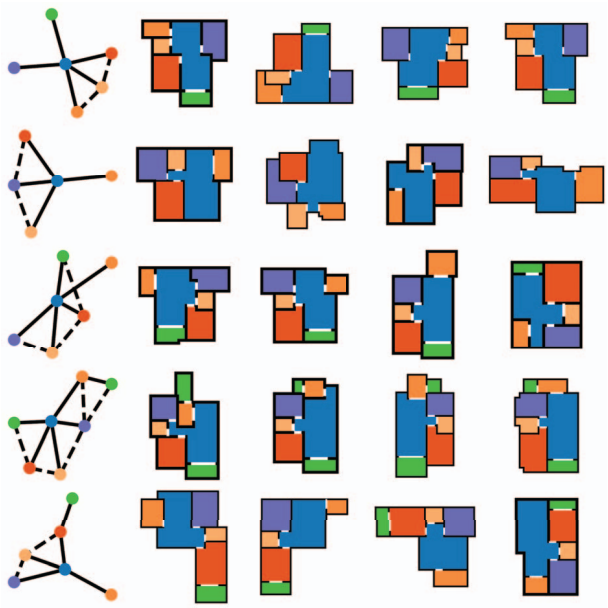


Figure 3. **Graph to image examples.** Each row represents an access graph (column 1) and several randomly selected corresponding floor plans semantic images (columns 2 - 5).

$$\text{SSIG}\left(p_1, p_2\right) = \frac{\text{mIoU}\left(X_1, X_2\right) + \left(1 - \text{nGED}\left(g_1, g_2\right)^{\gamma}\right)}{2}. \tag{7}$$

Coined as **SSIG** (**S**tructural **S**imilarity by **I**oU and **G**ED), 7 essentially computes a weighted average between a IoU-based and a nGED-based similarity metric. $\gamma \in \mathbb{R}^+$ is a weight that allows to tune between the relevance of nGED w.r.t. mIoU: the higher γ the *less* influence nGED and vice versa.[2] For example for RPLAN [26], the mIoU and nGED distributions are balanced when $\gamma = 0.4$. Specifically, balanced distributions here means that the IoU and nGED probability density functions have as much overlap as possible.

3.5. Efficient algorithm for ranking

Even for small graphs, GED is a compute-heavy measure, usually taking several seconds per floor plan pair. When **SSIG** is used for evaluation, pair mining, or direct floor plan retrieval on large datasets, the time constrain hugely limits practical usage. We propose an algorithm that drastically speeds up retrieval and finds structurally similar pairs relatively fast.

First, mIoU is computed for every combination of floor plan pairs in the dataset. Second, only the n (set between 50 and 100) best scoring pairs on mIoU for each floor plan identity are kept. Third, the **SSIG** is computed for each remaining pair. Fourth, the corresponding lists for each floor plan identity are ranked on **SSIG** in descending order.

The algorithm assumes that the best matches (highest **SSIG** scores) are found in the n-best mIoU pairs. In the next session, we empirically find that the assumption generally holds if n is large enough. In Subsec. 4.4, we show that for ranking RPLAN, the assumption holds for relatively small n. For a dataset with size N (usually between 10k and 100k), the algorithm is orders of magnitude faster than brute-forcing through it.

An assessment of the ranking algorithm and relation to floor plan retrieval systems are provided in the following section (Sec. 4).

4. Results and Evaluation

4.1. Data

In this paper, RPLAN [26] is used for analysis and evaluation. RPLAN is a large-scale dataset containing floor plan images with semantically segmented areas, consisting of >80k single-unit apartments across Asia. We further cleaned the dataset, where there were several apartments with rooms without any doors *i.e.* disconnected in terms of the topology. Moreover, apartments are removed that are not fully connected and/or for which our graph extraction algorithm cannot reliably compute the corresponding access graphs. The cleaned dataset contains ∼56k apartments. Furthermore, we found that RPLAN contains many (near) duplicates. For evaluation of the ranking algorithm (Subsec. 4.4), we remove the (near) duplicates to reduce clutter in the results. Duplicates are removed by a threshold on mIoU, $\tau_{\text{IoU}} = 0.87$. Duplicates are *not* removed when assessing the distributions and correlations of mIoU and nGED.

4.2. Distributions

To understand the relations between the pairwise similarity measures (mIoU and nGED) and how they relate to structural similarity, we start by investigating the probability density functions of the pairwise similarity measures (p_{IoU} and p_{GED}, respectively). p_{IoU} and p_{GED} are approximated by computing IoU and GED for over a million randomly sampled floor plan pairs. For brevity, we use "IoU" and "GED" instead of "mIoU" and "nGED".

IoU IoU is computed according to Eq. 3 and across all subdivision categories except for the "background". The histogram of the distribution is provided in Fig. 4 in the middle left plot in blue. The mean of the distribution is around 0.25 which means that on average approximately 25% of the pixels are overlapping. We also plot the IoU distribution for the 50 best-scoring examples on IoU per floor plan identity, shown in orange. As expected, compared to

[2] Note that nGED $\in (0, 1)$, hence increasing the power term decreases nGED.

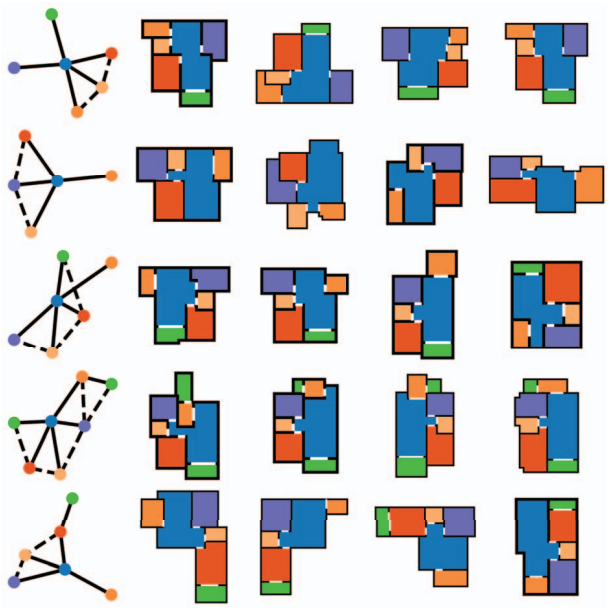


图3. 图到图像示例。每一行分别展示一个访问图（第1列）以及若干随机选取的对应楼层平面语义图像（第2至5列）。

$$\mathrm{SSIG}\left(p_1, p_2\right) = \frac{\mathrm{mIoU}\left(X_1, X_2\right) + \left(1 - \mathrm{nGED}\left(g_1, g_2\right)^{\gamma}\right)}{2}. \tag{7}$$

该方法被称为SSIG（基于IoU与GED的结构相似性），其本质是计算基于IoU的相似性度量与基于归一化广义编辑距离（nGED）的相似性度量之间的加权平均。其中，$\gamma \in \mathbb{R}^{+}$是一个权重参数，用于调节nGED相对于mIoU的重要性：γ越大，则nGED的影响力越强，反之亦然。[2]例如，在RPLAN[26]中，当γ =为0.4时，mIoU与nGED的分布趋于平衡。具体而言，此处的“分布平衡”是指IoU与nGED的概率密度函数具有尽可能大的重叠程度。

3.5. 用于排序的高效算法

即使对于小型图，GED也是一项计算密集型度量，通常每对楼层平面布局的计算需要数秒。当在大型数据集上使用SSIG进行评估、成对样本挖掘或直接的楼层平面布局检索时，时间约束极大地限制了其实际应用。为此，我们提出了一种能够显著加速检索过程并快速发现结构相似配对的算法。

首先，对数据集中每一对平面布局组合分别计算mIoU。其次，对于每个平面布局类别，仅保留mIoU得分位于50至100之间的最佳组合集合。第三，对每组重新剩余的配对。第四，针对每个楼层平面标识，相应的列表按SSIG值进行降序排序。

该算法假定最佳匹配（SSIG得分最高）位于n个最优的mIoU对中。在下一节中，我们通过实验发现，当n足够大时，这一假设通常成立。在第4.4节中，我们进一步表明，对于RPLAN排序而言，该假设在相对较小的n条件下亦成立。对于规模为N（通常介于1万至10万之间）的数据集，该算法的运行速度比穷举法快几个数量级。

对排序算法及其与楼层平面检索系统之间关系的评估将在下一节（第4节）中予以阐述。

4. 结果与评估

4.1 数据

在本文中，我们采用RPLAN[26]进行分析与评估。RPLAN是一个大规模数据集，包含带有语义分割区域的平面布局图像，涵盖亚洲地区的8万套单户型公寓（v15）。我们进一步对数据集进行了清洗，剔除了若干房间之间在拓扑结构上互不连通、且无任何门洞的公寓（v16）。此外，我们还移除了那些未完全连通，或者我们的图提取算法无法可靠地构建其访问图的公寓。经清洗后的数据集包含5.6万套公寓（v17）。进一步地，我们发现RPLAN中存在大量（近似）重复样本。为评估排序算法（第4.4节），我们按mIoU阈值0.87（v18–v19）剔除这些（近似）重复样本，以减少结果中的冗余。在分析mIoU与nGED的分布及相关性时，我们也同样剔除了重复样本（v20）。

4.2. 分布

为了理解成对相似性度量（mIoU 和 nGED）之间的关系及其与结构相似性的关联，我们首先考察这些成对相似性度量的概率密度函数（分别为 p_{IoU} 和 p_{GED}）。p_{IoU} 和 p_{GED} 通过计算超过一百万对随机采样的平面布局的 IoU 和 GED 来近似得到。为简便起见，我们用“IoU”和“GED”代替“mIoU”和“nGED”。

IoU 按照式（3）进行计算，且计算范围涵盖除“背景”之外的所有细分类别。其分布直方图如图4中左侧中间的蓝色曲线所示。该分布的均值约为0.25，这意味着平均约有25%的像素存在重叠。我们还绘制了按每张平面图标识计算的、IoU得分排名前50的样本的IoU分布，以橙色表示。正如预期的那样，与

[2]Note that nGED $\in (0, 1)$, hence increasing the power term decreases nGED.

the full IoU distribution, the mean is significantly shifted. The spike around 1.0 can be attributed to the many (near) duplicates in the dataset.

To give an idea of the diversity of floor plan samples, Fig. 4 (top row) highlights floor plans with different 'originality' scores. The originality is computed by the average IoU score for a given sample with all remaining samples. The more right (≈ 0.1) the more original *i.e.* seldom and the more left (≈ 0.4) the less original *i.e.* more frequent. Generally speaking, the originality positively correlated mostly with the complexity of the shapes of both the rooms and boundaries.

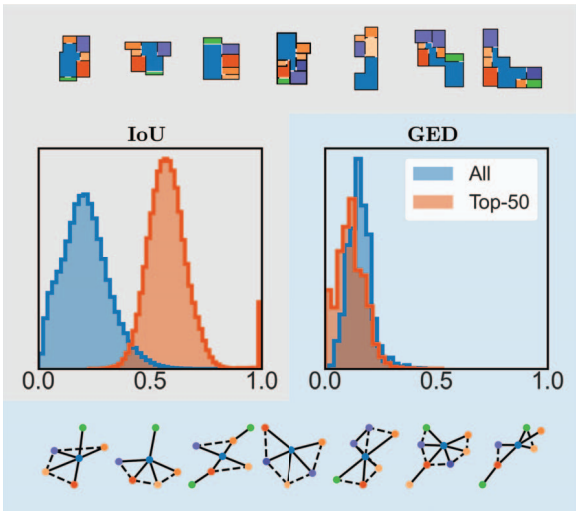


Figure 4. **IoU and GED distributions and originality.** The two plots in the center illustrate the distributions of IoU (left) and GED (right): blue depicts the distribution over all samples and orange only considers the top-50 *on IoU*. An indication of semantic image originality (based on the average IoU) is given on the top and one for the graph originality (based on the base graph occurrence) is given on the bottom: originality increases going from left to right.

GED nGED is computed according to Eqs. 5 and 6 and hence cost for node and edge deletions, insertions, and substitutions are kept the same. The histogram of the full distribution is illustrated in Fig. 4 in the middle right plot in blue. We also plot the GED distribution for the 50 best scoring examples *on IoU* per floor plan identity, shown in orange. Surprisingly, both full and top-50 distributions are nearly similar, revealing a weak (if at all) correlation between IoU and GED. The weak correlation is further investigated in the next section.

It is worth noting that there are only 50 unique GED values in our population of random pairs. The limited range of GED values can be attributed to:

Firstly, the amount of floor plan pairs that share the same underlying *base graph* – all possible graphs present in the dataset for which each pairwise combination is isomorphic:

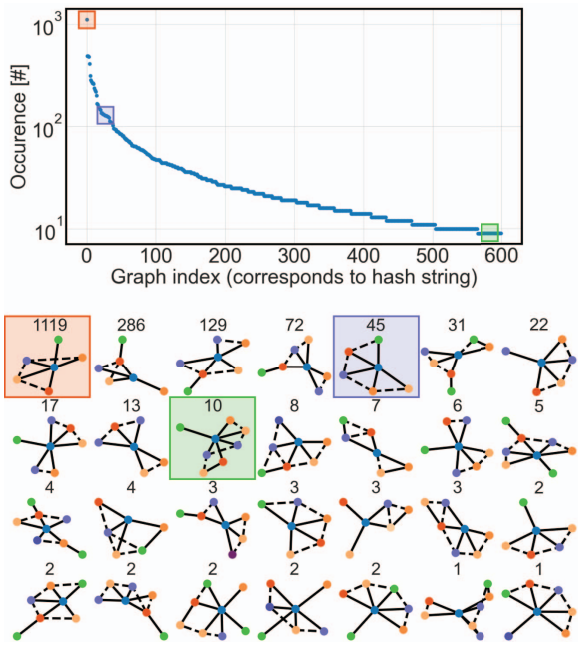


Figure 5. **Distribution of base graphs.** The plot on the top provides the distribution of base graphs (in descending order of occurrence) for base graphs that at least occur 9 times. On the bottom, the base graphs are plotted for several randomly picked floor plans. For some of the base graphs, we indicate the position in the distribution through the colored boxes.

g_1 is isomorphic to g_2 – is relatively large. To make our point, we sort all base graphs based on the number of corresponding floor plans and show the huge imbalance between base graph occurrences (see Fig. 5). For example, more than 25% of the data ($\approx$14k samples) correspond to the 100 most prominent base graphs only. Visualized in Fig. 4 (bottom row) and similar to the IoU originality scores, we score floor plans on graph originality as well. Graph originality is based on the number of corresponding floor plans for a given base graph: the more original the fewer floor plans per base graph. While the main patterns of graph originality are to be more closely investigated, on average the number of nodes as well as edges slightly increase with originality.

Secondly, revealed empirically by the graph originality distribution, the distribution of topology – the graph's structure – alone is small because the graphs are limited in size (number of connections) and order (number of subdivisions). Furthermore, the topological *types* are limited as well. Typically, access graphs are confined to be flower-like and thus centered around a single node. The central node is usually the living room and/or hallway.

Indeed, the majority of floor plans stem from a limited collection of base graphs, and only a few node and edge attributes are considered, leading to the ineffectiveness of measuring structural similarity solely based on graph similarity.

在完整的交并比（IoU）分布中，均值发生了显著偏移。在1.0附近出现的峰值可归因于数据集中存在的大量（近似）重复样本。

为展示平面布局样本的多样性，图4（顶行）展示了具有不同“原创性”得分的平面布局。原创性通过某一给定样本与其余所有样本之间的平均交并比（IoU）得分来计算：越靠右（≈ 0.1），原创性越高*i.e*，出现频率越低；越靠左（≈ 0.4），原创性越低*i.e*，出现频率越高。总体而言，原创性主要与房间及边界形状的复杂程度呈正相关。

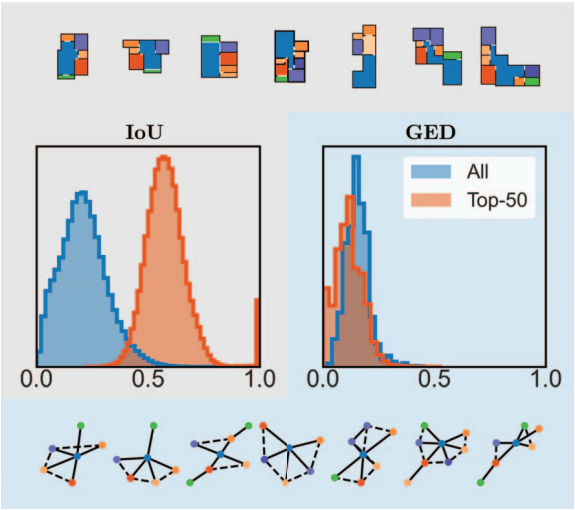


图4. IoU与GED的分布及原创性。居中的两幅图分别展示了IoU（左）与GED（右）的分布：蓝色表示所有样本的分布，橙色仅考虑前50名*on IoU*。顶部给出了基于平均IoU的语义图像原创性指标，底部则给出了基于基础图出现频率的图原创性指标；从左至右，原创性逐渐增强。

GED 和 nGED 按照式（5）和式（6）计算，因此节点与边的删除、插入及替换代价均保持一致。完整分布的直方图如图4中右上角的蓝色曲线所示。我们还绘制了按每个平面布局标识选取的得分前50个样本集*on IoU*的GED分布，以橙色表示。令人惊讶的是，完整分布与前50样本分布几乎相似，这表明IoU与GED之间几乎不存在相关性（即便存在也十分微弱）。这种较弱的相关性将在下一节中进一步探讨。

值得注意的是，在我们的随机配对样本中，仅有50个独特的GED值。GED值的取值范围较为有限，这可归因于：

首先，共享相同底层*base graph*的平面图对的数量——即数据集中所有可能的图中，每一对组合均彼此同构：

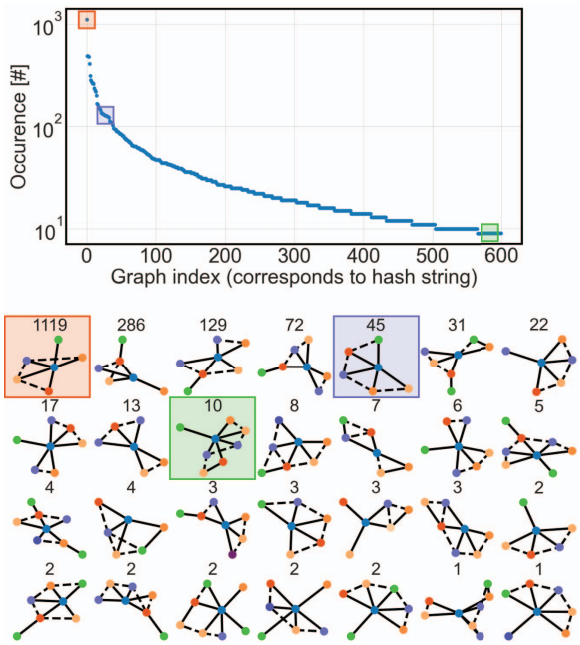


图5. 基础图的分布。上方的图表展示了至少出现9次的基础图的分布情况（按出现次数降序排列）。下方则针对若干随机选取的平面布局绘制了基础图。对于部分基础图，我们通过彩色方框标示其在分布中的位置。

g_1 与 g_2 同构——规模相对较大。为阐明这一观点，我们按对应楼面平面图的数量对所有基础图进行排序，并展示了基础图出现频次之间的显著不均衡（见图5）。例如，超过25%的数据（≈1.4万样本）仅对应于排名前100的基础图。如图4（底部一行）所示，与IoU原创性评分类似，我们也基于图的原创性对楼面平面图进行评分。图的原创性以给定基础图所对应的楼面平面图数量为依据：原创性越高，每个基础图对应的楼面平面图数量越少。尽管图的原创性模式有待进一步深入研究，但平均而言，随着原创性的提高，节点数和边数均略有增加。

其次，从图的原创性分布所揭示的实证结果来看，仅就拓扑结构（即图的结构）而言，其分布范围较小，这是因为这些图在规模（连接数量）和阶数（子划分数量）上均受到限制。此外，拓扑特征也较为有限。通常，接入图呈现出类似“花状”的结构，因而以单个节点为中心。该中心节点通常是客厅和/或走廊。

事实上，大多数平面布局均源自有限的一组基础图，且仅考虑少数节点和边的属性，这导致仅基于图相似性来度量结构相似性的方法效果不佳。

4.3. Correlations

The correlation between IoU and GED is further investigated. The 2D density map of IoU and GED is depicted in Fig. 6 and plotted as a heatmap. We define a positive correlation when increments in visual similarity *i.e. higher* IoU result in increments in the graph-based similarity *i.e. lower* GED. Effectively we measure the correlation between IoU and *negative* GED. The heatmap shows for increasing IoU ranges that GED distributions shift towards lower values, hence the correlation between IoU and GED is positive. To be precise, the Pearson correlation coefficient is 0.503 between mIoU and -nGED, hence GED and IoU are moderately correlated. However, the overlapping regions between GED distributions for different IoU ranges are often large. Specifically, IoU and GED oppose each other 38% of the time – opposition here means that a higher IoU results in a higher GED or that a lower IoU results in a lower GED. Several examples in which IoU and GED oppose each other are given in Fig. 7.

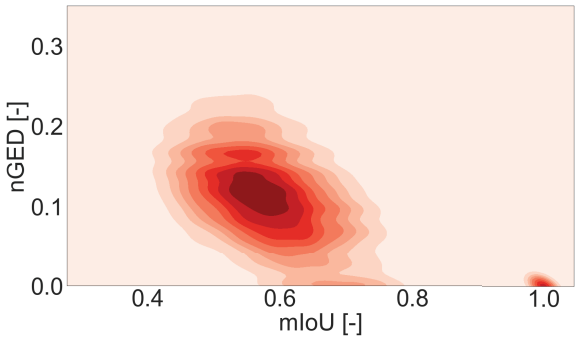


Figure 6. **2D density map of IoU and GED as heatmap.** A moderate correlation (0.503) between IoU and -GED is found. The distribution is, however, widely spread indicating that IoU and GED often fail to predict one another with confidence.

From the triplets shown in Fig. 7, many characteristics of IoU and GED rise to the surface. For an anchor, we define pos@IoU as the positive (IoU high) and neg@IoU as the negative query (IoU low). The same holds for pos@GED (GED low) and neg@GED (GED high). Most prominent is the sensitivity of IoU w.r.t. the overall shape of the building: in all cases, the overall shape between the anchor and the pos@IoU are nearly identical and slightly different for the pos@GED. On the other hand, the number of rooms is often different between the anchor and pos@IoU while (nearly) always equivalent between the anchor and pos@GED: placing or removing rooms is detrimental for GED but in many cases does not hurt IoU so much. Another difference between pos@IoU and pos@GED is the fact that changes in the room shapes – especially to the central space – are *not* penalized by GED yet detrimental for IoU. The examples reveal the difficulty of measuring the structural similarity of floor plans. Apparently, both measures independently cannot fully grasp structural similarity.

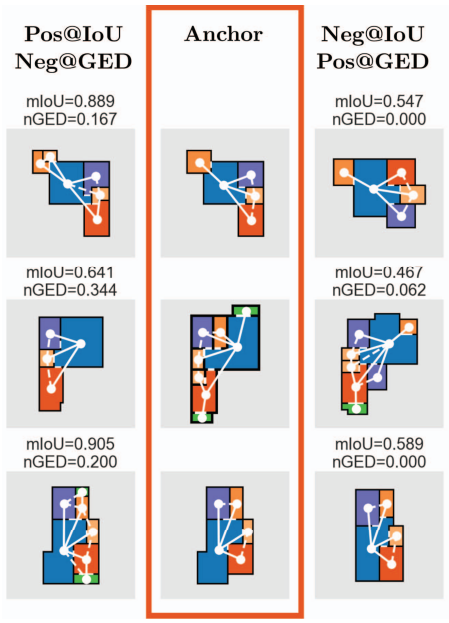


Figure 7. **Negative examples between IoU and GED on RPLAN.** IoU is sensitive to the scales and shape of the room while GED captures the connections between the rooms.

4.4. Floor plan retrieval

We evaluate our ranking algorithm provided in Subsec. 3.5 and qualitatively compare the retrievals to independent IoU- and GED-based rankings, and to retrievals based on a deeply learned metric [13].

We start by finding a proper value for n – the number of samples (when ranked on IoU) considered for further **SSIG** assessment. In Fig. 9, the distributions for IoU, GED, and **SSIG** are provided for various values of n. In our observation, for relatively large values of **SSIG** ($>$0.7) every linear increase in n adds fewer samples every time. We found for RPLAN that values of $n >= 50$ seldom change the top-10 on **SSIG**. Compared to measuring **SSIG** on the whole dataset, **SSIG** only needs to be computed 50 times. We believe that IoU, at least in the case of RPLAN, is a proper first step to find correspondences in structural similarity coarsely, while GED is a proper help to measure structural similarity on a finer level. Hence, we refer to our ranking method as a visually-guided graph edit distance.

Our algorithm is compute-efficient w.r.t. GMNs. Depending on the size and originality of the floor plan's corresponding access graphs, our algorithm takes between 1 and 5 s on a single CPU per unseen sample to fully rank it. Compared to LayoutGMN, the same computation takes approximately 1500 s. Hence, our algorithm is compute-efficient w.r.t. GMNs.

Randomly picked retrievals (top 10) of our method are

4.3. 相关性

我们进一步研究了IoU与GED之间的相关性。图6展示了IoU与GED的二维密度分布，并以热图形式呈现。当视觉相似度*i.e.*、*higher*的IoU值增加时，基于图的相似度*i.e.*、*lower*的GED值也随之增加，我们将其定义为正相关。实质上，我们是在量化IoU与*negative*GED之间的相关性。热图显示，随着IoU取值范围的增大，GED的分布逐渐向较低值方向偏移，因此IoU与GED呈正相关。具体而言，mIoU与−nGED之间的皮尔逊相关系数为0.503，表明GED与IoU存在中等程度的相关性。然而，不同IoU区间对应的GED分布之间往往存在较大的重叠区域。特别地，在38%的情况下，IoU与GED呈现出相反的变化趋势——此处的“相反”是指较高的IoU对应较高的GED，或较低的IoU对应较低的GED。图7中列举了若干IoU与GED呈现相反变化的具体实例。

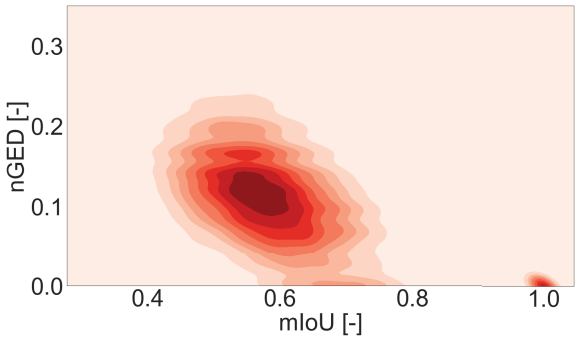


图6. IoU与GED的二维密度分布热图。结果显示，IoU与-GED之间存在中等程度的相关性（0.503）。然而，数据分布较为分散，表明IoU与GED在预测彼此时往往缺乏较高的置信度。

从图7所示的三元组中，IoU与GED的诸多特性得以显现。对于一个锚框，我们定义pos@IoU为正样本（IoU较高），neg@IoU为负样本（IoU较低）；同样地，pos@GED指GED较低的正样本，neg@GED指GED较高的负样本。最为显著的是，IoU对建筑物整体轮廓的敏感性：在所有情形下，锚框与pos@IoU之间的整体轮廓几乎完全一致，而与pos@GED则存在细微差异。另一方面，锚框与pos@IoU之间的房间数量往往不同，但与pos@GED之间的房间数量却几乎总是相等：增减房间会显著增大GED，但在许多情况下对IoU的影响并不明显。pos@IoU与pos@GED之间的另一区别在于，房间形状的变化——尤其是对中心空间的调整——会被GED所惩罚，却对IoU产生不利影响。这些实例揭示了度量建筑平面结构相似性的难度。显然，这两种度量指标各自独立地能够—

not fully grasp structural similarity.



图7. 在RPLAN数据集上，IoU与GED之间的负例示意图。IoU对房间的尺度和形状较为敏感，而GED则能够捕捉房间之间的连接关系。

4.4. 平面图检索

我们评估了第3.5节中提出的排序算法，并从定性角度将其检索结果与基于IoU和GED的独立排序方法，以及基于深度学习度量的学习型检索方法[13]进行了比较。

我们首先确定n的合理取值——即在按IoU排序后用于进一步SSIG评估的样本数量。图9展示了针对不同n取值时，IoU、GED和SSIG的分布情况。根据我们的观察，当SSIG取值相对较大（$>\geqslant 0.7$）时，每次线性增加n所新增的样本数量逐渐减少。对于RPLAN而言，我们发现n >==50时，SSIG排名前10的样本几乎不会发生变化。与在整个数据集上计算SSIG相比，此时只需计算50次即可。我们认为，在RPLAN的情形下，至少就粗粒度的结构相似性匹配而言，IoU是恰当的初始步骤；而GED则有助于在更精细的层次上度量结构相似性。因此，我们将所提出的排序方法称为“视觉引导的图编辑距离”。

与GMN相比，我们的算法在计算效率上更具优势。根据平面布局所对应的访问图的规模及其独特性，对于每个未见过的样本，我们的算法在单个CPU上完成其完整排序所需的时间介于1至5秒之间。相比之下，LayoutGMN完成相同计算则需约1500秒。因此，与GMN相比，我们的算法在计算效率上更具优势。

我们方法的随机抽取检索结果（前10条）为



Figure 8. **Retrieval results of our method.** The most left is the query and to the right are the retrievals. The red and blue transparent boxes indicate the IoU and GED ranks, respectively.

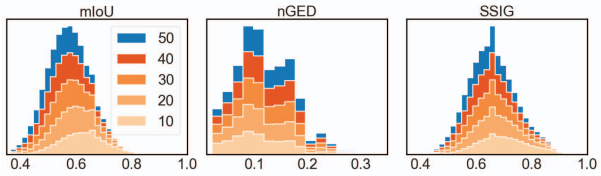


Figure 9. **IoU, GED, and SSIG distributions for various** n. The different colors indicate different n. Particularly interesting is the fact that the mean of both IoU and SSIG drastically decreases for increasing n.

shown in Fig. 8 and compared against IoU- and GED-only ranks. In red and blue are IoU and GED ranks, respectively. Not surprisingly the **SSIG**, IoU, and GED ranks are quite different which can be attributed to the fact that the correlation between IoU and GED is only weakly positive (see Subsec. 4.3). Compared to IoU- and GED-only retrievals, it can be seen that **SSIG** indeed compensates for 'failures' in IoU and GED. For example in the last row of Fig. 8, the first 4 retrieval results, even though having a low IoU value, are compensated by GED, while the 5th retrieval is compensated by IoU, even though having a low GED value. Similar compensation for failures can be found back in the retrievals of deeply learned metrics, see [13, 4, 8].

5. Conclusion and future work

In this work, we showed that image- as well as graph-based similarity metrics alone are likely not enough to robustly measure structural similarity between floor plans. We overcome the shortcomings by combining IoU and GED into one metric, called **SSIG**. We qualitatively showed that a floor plan database which is ranked on **SSIG** shares similar characteristics to results of state-of-the-art methods that use GMNs to compute the ranking.

This work is however limited to investigating IoU and GED and does not consider other well-known image- and graph-based similarity metrics. Moreover, we did not conduct any user study to rigorously assess our claims further. In future studies we hope to address both limitations.

The formulation of similarity as defined in Eq. 1 is setup in a task-agnostic manner - the formulation addresses floor plan similarity as a *single* overall measure. In many cases, however, a more realistic way to setup similarity might be dynamic (*e.g.*, based on categories), hence, allowing for different similarity scores across multiple use cases whether used in floor plan tailored search engines, evaluation of floor plan generation, etc. Our aim is to work on such formulations in the future.

We believe that **SSIG** will pave a way towards improved data-driven research on floor plans, such as for providing a more natural objective to train deep metric learning frameworks that do not necessarily require information sharing across branches.

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图8. 本方法的检索结果。最左侧为查询图像，右侧为检索结果。红色和蓝色的半透明框分别表示IoU排名和GED排名。

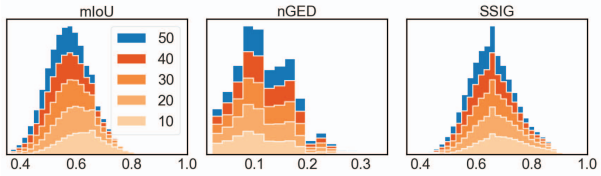


图9 不同v0下的IoU、GED和SSIG分布。不同颜色代表不同的v1。尤其值得注意的是，随着v2的增大，IoU和SSIG的均值均显著下降。

如图8所示，并与仅基于IoU和仅基于GED的排序进行了对比。红色和蓝色分别表示IoU和GED的排序结果。不出所料，SSIG、IoU和GED的排序结果存在显著差异，这可归因于IoU与GED之间的相关性仅为弱正相关（参见第4.3节）。与仅基于IoU或仅基于GED的检索结果相比，可以看出SSIG确实能够弥补IoU和GED在某些情况下的不足。例如，在图8的最后一行中，前4个检索结果尽管IoU值较低，但通过GED得到了补偿；而第5个检索结果尽管GED值较低，却通过IoU得到了补偿。类似的补偿机制同样存在于深度学习度量的检索结果中，详见文献[13, 4, 8]。

5. 结论与未来工作

在本研究中，我们表明，仅依靠基于图像或基于图的相似性度量，可能不足以稳健地衡量楼层平面图之间的结构相似性。为此，我们将交并比（IoU）与图编辑距离（GED）相结合，构建了一种名为SSIG的综合度量指标，从而克服了上述不足。定性分析结果表明，按照SSIG排序的楼层平面图数据库，在特征上与现有最先进方法的输出结果具有较高的相似性。

使用广义匹配网络（GMNs）进行排序计算。

然而，本研究仅限于对IoU和GED的考察，并未考虑其他广为人知的基于图像与图的相似性度量。此外，我们尚未开展任何用户研究以进一步严格验证我们的结论。在后续研究中，我们期望能够同时解决上述两项局限性。

如式1所定义的相似性度量采用了一种与任务无关的设定方式——该度量将平面布局的相似性视为一个整体性的指标。然而，在许多情况下，更为现实的相似性建模方式可能是动态的（例如，基于类别），从而在多种应用场景下允许不同的相似性评分，如用于定制化的平面布局搜索引擎、平面布局生成的评估等。我们的目标是在未来进一步研究此类建模方法。

我们相信，SSIG将为基于数据的平面布局研究开辟新途径，例如为深度度量学习框架提供更为自然的优化目标，而这些框架并不一定需要在不同分支之间进行信息共享。

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